

Xiuwei Zhang

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EDUCATION

2023.09 - 2025.01

University of Bristol

MSc, Optical Communication and Signal Processing

Core Courses: Optical Communication Systems, Optoelectronic Devices

Thesis: Covert Quantum Ranging in High-Noise Environments (Supervisor: Prof. John Rarity)

2020.09 - 2023.06

University of Central Lancashire

BEng (Hons) Electronic Engineering (2:1)

Core Courses: Computer Vision, Digital Image Processing, Communication Principles

Undergraduate Thesis: Robust Object Detection and Tracking in Cluttered Scenes

RESEARCH & PROJECT EXPERIENCE

Covert Quantum Ranging System Development

University of Bristol, 2024 | Supervisor: Prof. John Rarity (Micius Quantum Prize laureate)

- Developed an automated single-photon ranging system based on energy-time correlation, achieving $\pm 0.3\text{m}$ accuracy over 150m and strong noise suppression.
- Designed MATLAB-based data acquisition and multi-channel signal processing, enabling 24-hour unattended operation.
- Benefited from direct mentorship in a leading quantum/optical communication group.

Silicon Photonic Device Simulation

- Modelled SOI waveguides, calculated modal cutoff, dispersion and nonlinear parameters using COMSOL.
- Supported advanced photonic device design with both theoretical and simulation approaches.

Robust Object Detection and Tracking in Cluttered Scenes

Undergraduate Thesis, UCLan | Supervisor: Prof. Bogdan Matuszewski

- Built and benchmarked a SIFT/SURF-based real-time object detection and tracking system in MATLAB for complex visual environments.

Communication System Modelling & MATLAB Simulation

- Analysed channel capacity, BER performance of QPSK/16QAM/AWGN/MIMO systems using MATLAB and built models for signal transmission and error correction.

TECHNICAL SKILLS

- Programming/Simulation:** MATLAB, Python, Verilog, COMSOL
- Technical:** Optical/quantum communication, single-photon detection and LIDAR
- Languages:** English (fluent) and Mandarin (native)
- Soft Skills:** Teamwork, independent research and project management

KEY STRENGTHS

- All major reports and thesis written and presented in English.
- Highly adaptable, with practical, hands-on training in a leading quantum communication lab.
- Recognised by Prof. John Rarity for meticulous and efficient work.